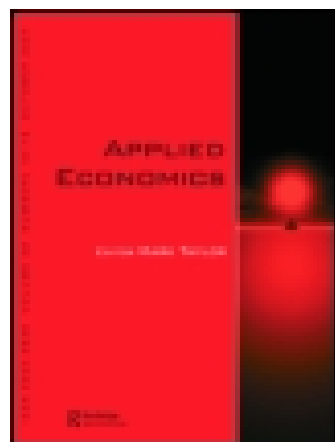


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Common dynamics of nonenergy commodity prices and their relation to uncertainty

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The purpose of this article is to improve the empirical evidence on commodity prices in various dimensions. First, we attempt to identify the extent of comovements in 44 monthly nonenergy commodity price series in order to ascertain whether the increase in comovement is a recent term phenomenon. Second, we attempt to determine the role of uncertainty in determining comovements among nonenergy prices in the short run. We diagnose the overall comovement using a dynamic factor model estimated by principal components. A factor-augmented vector autoregressive approach is used to assess the relationship of fundamentals, financial and uncertainty variables with the comovement in commodity prices. We find a greater synchronization among raw materials since December 2003. Since that date, uncertainty has played an important role in determining short-run fluctuations in nonenergy raw material prices.

Keywords: commodity prices; comovement; factor-augmented vector autoregressive models

JEL Classification: E30; F00

1. Introduction

One interesting characteristic of raw material prices is their tendency to comove in response to the same shock. Thus, prices which should apparently not be correlated have a common evolution in time. For Pindyck and Rotemberg (1990), there are macroeconomic variables such as the industrial production of developed countries, interest rates and the exchange rate, which influence a large group of raw materials, since they affect the current market as well as the short-term expectation of demand and future supply. Pindyck and Rotemberg (1990) found that commodity price comove in excess of what can be

explained by these variables. This evidence casts a shadow of doubt over the competitive formation of prices in the commodity market and offers the possibility of studying the role of speculation in this market.

Recently, and in view of the significant increase in raw material prices in the period between 2002 and 2008 (see Fig. 1), researchers have once again taken up the debate on the importance of the basic macroeconomic variables which may affect comovement with regard to the role of speculation by index investors or so-called financialization of commodity markets (see, e.g. Irwin *et al.*, 2009; Tang and Xiong, 2012). An intense debate among scholars started with a hedge fund manager's testimony, that of Mr

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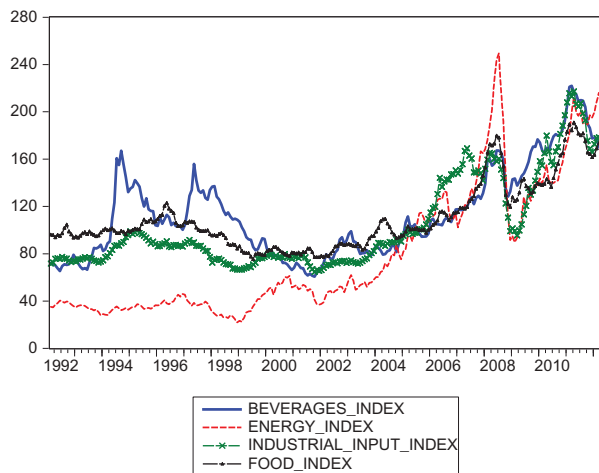


Fig. 1. Indices of market prices for nonfuel (beverage food and industrial inputs) and fuel (energy) commodities (2005 = 100, in terms of US dollars)

Source: Author's graph based on commodity indices from International Monetary Fund (IMF) Primary Commodity Prices.

Michael W. Masters, at the US Senate. Masters (2008) drew attention to the fact that the increase in speculative activity in commodity markets, due to the arrival of new financial traders like corporate and government pension funds, modified price determination in the commodity market. According to Masters (2008), assets allocated to commodity indices by institutional investors had risen from 13 billions at the end of 2003 to 260 billions as of March 2008, driving commodity prices higher.

The aim of this article is to improve the empirical evidence on commodity price comovement in several dimensions. First, we attempt to determine the extent of comovements in a larger set of commodities; in particular, we analyse 44 monthly nonfuel commodity price series. The aim is to ascertain whether the links among commodity prices, or comovement, have increased since the end of 2003, as suggested by the promoters of the financialization hypothesis (Masters, 2008; Tang and Xiong, 2012; among others). Second, we attempt to determine the role of uncertainty as a potential driver of nonenergy comovement in the short term. The extent to which this comovement is driven by macroeconomic fundamentals and financial variables is also examined, in order to analyse their relative importance. While we evaluate the effect of uncertainty on commodity comovement, we add to the recent business cycle literature on the macroeconomic effects of 'uncertainty' shocks (Bloom, 2009; Bernanke, 2012; Baker et al., 2013).

We use monthly nonfuel commodity price data from January 1992 until December 2012 and proceed in two steps: first, the use of a large number of commodities and

data from different sectors allows us to use a dynamic factor model (DFM) approach, and to estimate it through principal components. We diagnosed the overall comovement and confirmed the presence of one latent factor-driving commodity prices with the information criteria suggested by Bai and Ng (2002). While we were evaluating the importance of this common factor for the variance of nonfuel commodity prices before and after the end of 2003, we found a very interesting fact: the variance of commodity prices explained by the common factor jumped from 9% in the first sub-period, February 1992 to November 2003, to 23% in the second sub-period, December 2003 to December 2012. In terms of the scope of the analysis, we present a 'before and after' perspective by looking at two periods: before and after December 2003. Second, we use a factor-augmented vector autoregressive (FAVAR) approach, proposed by Bernanke et al. (2005), to study if the comovement of commodity prices is affected in the same way by macroeconomic and speculative variables, as well as the uncertainty variable in the two sub-periods.¹

Our study provides an understanding of the role of uncertainty and fundamental variables that drive comovement in commodity prices in these two periods. Therefore, our contributions to the literature on comovement in commodity prices are twofold. First, by considering two different periods, we check whether comovement is affected in the same way by fundamental and speculative variables, irrespective of time. It is our aim to ascertain which variables – fundamental, financial or uncertainty – have a greater influence on the common behaviour of nonfuel commodity prices.

Second, we examine the importance of market uncertainty in the comovement of a range of nonfuel commodity prices in the short term. The closest paper to ours is Byrne et al. (2013), although the two approaches differ in several aspects. First, we focus on the short-term relationship between comovement in nonfuel commodity prices and macroeconomic and financial variables on a monthly instead of annual basis. The use of higher frequency data allows us to focus on fluctuations due not only to fundamentals, such as the real exchange rate and the world demand, but also to speculative or financial causes. We believe that the effects of uncertainty over commodity prices are better analysed in a short-term context. Second, we apply a previously unused variable in the study of uncertainty in commodity prices: the Chicago Board Options Exchange Volatility Index (VIX). This variable reflects a market estimate of future volatility based on the weighted average of the implied volatilities for a wide range of options. Since uncertainty is forward-looking, the variable VIX should be a better measure of uncertainty,

¹ The Chow test rejects the null of no structural break in December 2003 for the regression of the common evolution on nonenergy commodity prices against macroeconomic and financial variables.

given that it takes into account the expected volatility in the future market. Third, we split the sample into two parts in order to analyse the differences along comovement before and after the end of 2003 more efficiently. We believe that the sharp increase in comovement after that date merits different treatments in the FAVAR model. Fourth, the estimation properties regarding consistency of the factor comovement are better achieved as we do not concentrate on a few commodities, as Byrne *et al.* (2013) do, but try to include as many as possible. Finally, we add two more variables to the FAVAR – a stock market index variable (MSCI World Index), and the real US exchange rate – as they affect comovement in the short term according to the literature (e.g. Vansteenkiste, 2009; Coleman, 2012).

The rest of the article is organized as follows. Section II reviews the methodology used in the study. Section III shows the main results. The conclusions are presented in Section IV.

II. Methodology

The FAVAR model

In the first part of this section, we provide a summary of the FAVAR model that we use in the empirical section. For additional details, see Bernanke *et al.* (2005).

DFMs suggest that the information from a large number of time series can be summarized by a relatively small number of common factors plus idiosyncratic noises. In this context, we extract a factor that represents comovements in commodity prices. We would like to assess for the responses of this factor to common macroeconomic and speculative shocks. For this purpose, we propose a FAVAR approach along the lines of Bernanke *et al.* (2005). The model is summarized in the following two equations:

$$\begin{bmatrix} F_t \\ X_t \end{bmatrix} = \phi(L) \begin{bmatrix} F_{t-1} \\ X_{t-1} \end{bmatrix} + v_t \quad (1)$$

$$Y_t = \Lambda F_t + u_t \quad (2)$$

where $Y_t = (Y_{1t}, \dots, Y_{Nt})'$ is the $N \times 1$ vector of observed variables (the inflation of each nonenergy commodity in our case); $\Lambda = (\Lambda_1', \dots, \Lambda_N')$, $N \times k$ is the factor loading matrix; F_t is the $k \times 1$ vector of common factors; u_t is the $N \times 1$ vector of idiosyncratic noises; X_t is the $p \times 1$ vector of macroeconomic and financial variables; $\phi(L)$ is the $(p+k) \times (p+k)$ matrix of lag polynomials and $v_t \sim (0, \sum_v)$ is the $(p+k) \times 1$ vector of error terms in the FAVAR specification.

Note that in Equation 1 the latent variable (F_t) summarizes the developments of nonenergy commodity

prices. Therefore, the factors F_t represent the common pattern of commodities. In this context, Equation 1 represents the joint dynamics of (F_t, X_t) , and we call it FAVAR.

We cannot directly estimate Equation 1 because the factors (F_t) are unobservable; therefore, we need to construct the factors beforehand. In our case, the DFM given by Equation 2 gives the inflation of each nonenergy commodity (labelled as Y_t) driven by a latent component F_t that is common to all series and an idiosyncratic AR component u_t . The matrix Λ represents the loading of the common factors on to series. Each element of the error or idiosyncratic term, u_t , contains the dynamics specific to each commodity price inflation, although it is assumed to be weakly correlated. The factors may also follow an AR processes.

With regard to the estimation, we consider the two-step method favoured by Bernanke *et al.* (2005), in which the factors are extracted prior to estimation of the FAVAR. The use of a large number of commodities allows us to estimate Equation 2 through principal components. With this, we assume that the weighted averages of the idiosyncratic disturbances will converge to zero by the weak law of large numbers, so linear combinations of the observed series are consistent estimators of the common factors. Consistency of the static principal components estimator has been demonstrated by Stock and Watson (2002) when both the number of series N and the time dimension T converge to infinity. Due to this consistency result, we treat F_t as observed for inference purposes.

Data definition

In this section, we describe the data used in the factor model, related to commodity prices and the macroeconomic and financial variables used in the FAVAR model.

For the dynamic factor analysis, we used 44 monthly nonfuel commodity price series from the period February 1992 to December 2012. Monthly series of commodity prices were obtained from the International Monetary Fund (IMF) Primary Commodity Prices database. We included in our study the raw materials that were available in the following categories: food, beverages, agricultural raw material and metals. A summary of the commodities used in this study is shown in Appendix 1.

Commodity prices are log differentiated and standardized, prior to the factor extraction by principal components. We follow Stock and Watson (2011) in this respect.

With the FAVAR model, we attempt to determine the extent to which this common factor is driven by macroeconomic fundamentals, or whether uncertainty plays a major role. For this purpose, we select the most important macroeconomic variables used in the literature of comovements and combine them with our proxy for uncertainty. Here, we add the role of uncertainty as a potential determinant of commonalities in commodity prices. Specifically, we use the VIX which reflects a market

estimate of future volatility. To the best of the authors' knowledge, this variable has not been used before in the analysis of commodity prices. An additional financial variable we used in this study is the Morgan Stanley Capital International (MSCI) World Index, which includes a large collection of world stocks in the developed markets. We use this variable as a proxy of the financial market condition. From now on we called this variable world stock index. The remaining variables are macroeconomic fundamentals summarized as follows:

- (1) The United States exchange rate. We use the US real effective exchange rate. We expect that the decline in the real effective exchange rate may have added momentum to the upward commodity price movement.
- (2) United States real interest rate, proxied in our analysis by the US 3-Month Certificate of Deposit. We expect that lower rates of return on bonds will increase the speculative demand for commodities, and hence raise their price further.
- (3) World demand, proxied in our analysis by the World Industrial Production.
- (4) Supply shocks, proxied in our study by the energy index of the IMF. The increase in oil prices may explain the contemporaneous increase in other commodities, such as food products, through two different channels. First, higher energy prices cause upsurges in the production cost which impacts on the rest of the final price of commodities. Second, the increased biofuel demand may lead to a reduction in food supply devoted to final consumption. The last two channels are more associated to long-run effects, since raw material supply is highly inelastic. In the short run, if the financialization hypothesis prevails, we would expect that energy and nonenergy prices will move in the same direction, given the speculative trade on commodities. The energy index is composed of natural gas and coal, apart from oil.

Graphical representations of the variables are shown in Appendix 2.

III. Empirical Results

In this section, we focus first on the specification of the DFM and the analysis of the factor and loadings estimated by principal components. The estimation of the unobserved factors is the first step, since we estimate the FAVAR through the two-step approach of Bernanke *et al.* (2005). Afterwards, we move on to the estimation of the FAVAR for both periods and compare their results.

Dynamic factor model

We identify the factor structure using the information criteria proposed by Bai and Ng (2002). Bai and Ng (2002) solve the optimization problem that comes from minimizing the sum of squared residuals (divided by NT), defined by:

$$V(k) = (NT)^{-1} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \lambda_i^k F_t^k)^2 \quad (3)$$

where the super index k in $\lambda_i^k F_t^k$ denotes that k factors are considered. To determine the true number of factors r (with $r \leq k_{\max}$), Bai and Ng (2002) formulate the following information criteria:

$$IC(k) = \ln \left(V(k, \widehat{F}^k) \right) + kg(N, T) \quad (4)$$

where $V(k, \widehat{F}^k)$ is the average residual variance when k factors are assumed and $g(N, T)$ is a penalty function. Bai and Ng (2002) suggested three penalty functions to be taken into account for principal component estimation as follows:

$$g_1(N, T) = \frac{N+T}{NT} \ln \left(\frac{NT}{N+T} \right) \quad (5)$$

$$g_2(N, T) = \frac{N+T}{NT} \ln(c_{NT}^2) \quad (6)$$

$$g_3(N, T) = \frac{\ln(c_{NT}^2)}{c_{NT}^2} \quad (7)$$

where

$$c_{NT} = \min \left\{ \sqrt{N}, \sqrt{T} \right\}.$$

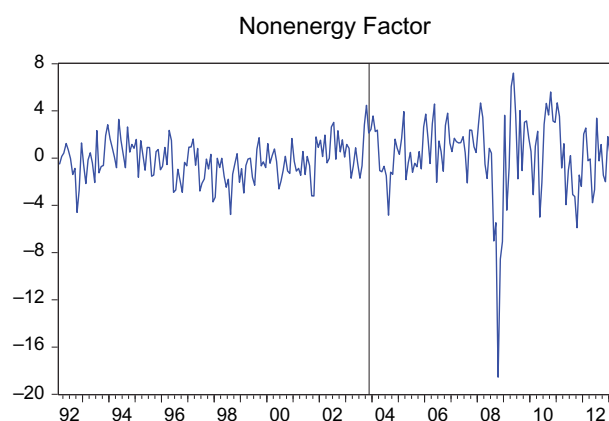
In our case, the three criteria agree that there is one common factor in the data, as can be seen in Table 1. Accordingly, we estimate one common factor, which we name comovement, for the entire sample and also for the two periods we want to analyse: pre-December 2003 and post-December 2003.

Figure 2 plots the first estimated common factor, which we call comovement in nonfuel commodity prices. According to this factor, commodity price booms and busts tend to be relatively short-lived, and the factor is more volatile after the end of 2003. Importantly, the variance explained by the factor increases significantly between subsamples. When estimating the first principal component for the period between February 1992 and November 2003, the proportion of the variance of commodity prices explained by the factor is only 9%. However, when the same procedure is performed for the

Table 1. Number of factors estimated using Bai and Ng (2002) information criteria

Sample	Dates	Number of obs.	IC_1	IC_2	IC_3
Full	1992:2–2012:12	251	1	1	1

Note: All estimates use $N = 44$ series.

**Fig. 2. Common factor for nonenergy commodity prices**

period between December 2003 and December 2012, the variance explained increases to 23%. This would suggest that after the end of 2003, nonfuel commodity prices have become more synchronized, or that fluctuations are higher.

Figure 3 shows the commodities with factor loadings above 0.10 in both periods. Although loadings in the first sub-period are larger for four commodities (greater than 0.30), fewer raw material prices reach 0.10 in their factor loading. In contrast, for the second sub-period, there are more commodities with loadings above 0.10. The correlation of the factor with each of the prices is greater in most of the commodities for the second sub-period. It is worth noting that some commodities greatly increased their relationship with the factor in the second period. Sugar, Rubber and Zinc, for example, had a factor loading of only 0.123, 0.105 and 0.128, respectively, in the first period. After December 2003, however, their loadings went up to 0.21, for all three commodities. The relationship between commodity prices and the factor increased for edibles in the IMF category of ‘food’ such as fish and beef and in ‘beverages’ such as tea and coffee. In contrast, the factor loading decreased for ‘raw materials’ related to the group vegetable oils and protein meal.

	1992:2–2003:11	
	Loadings	Average correlation/ Global factor
Soybeans	0.393	0.803
Soybean oil	0.353	0.721
Maize	0.319	0.650
Soybean meal	0.311	0.634
Sunflower/Safflower Oil	0.300	0.612
Barley	0.288	0.588
Wheat	0.229	0.469
Nickel	0.221	0.451
Rapeseed oil	0.195	0.398
Tin	0.186	0.379
Copper	0.175	0.358
Cotton	0.168	0.343
Lead	0.168	0.343
Sugar	0.123	0.252
Zinc	0.128	0.261
Rubber	0.105	0.214

	2003:12–2012:12	
	Loadings	Average correlation/ Global factor
Soybean oil	0.266	0.803
Copper	0.238	0.720
Soybeans	0.231	0.698
Tin	0.225	0.678
Coarse	0.219	0.661
Wool Fine	0.217	0.654
Rubber	0.213	0.645
Sugar	0.211	0.639
Zinc	0.210	0.633
Rapeseed oil	0.209	0.631
Maize	0.207	0.625
Barley	0.207	0.624
Nickel	0.200	0.605
Lead	0.191	0.577
Soybean meal	0.183	0.553
Cotton	0.180	0.544
Wheat	0.168	0.509
Other Milds of Coffee	0.165	0.498
Coffee Robusta	0.158	0.478
Cocoa beans	0.135	0.409
Sawnwood	0.127	0.384
Fish	0.123	0.371
Olive Oil	0.121	0.364
Sugar Free Market	0.120	0.362
Groundnuts	0.117	0.353
Aluminium	0.110	0.332
Sugar US market	0.110	0.331
Uranium	0.103	0.312

Fig. 3. Variance shares explained by the common factor and correlation for the two sub-periods. The table shows only commodities with a loading greater than 0.10

As an additional exercise in this section, we present the estimation results of an extension DFM of the general model by evaluating the common evolution of commodity prices in small categories of related raw materials. Therefore, we estimate, through the Kalman filter, individual DFMs for each of the following categories: cereals, beverages, vegetable oil and protein meals, agricultural raw materials, metals and meat and seafood. A summary of the commodities under each category is given in Appendix 1.

The resulting factors estimated from each one of these models are presented in Fig. 4. As it can be seen from simple observation of the graph, the increase in volatility of the common factor that encompasses the whole data set is not obvious in the case of commodity categories. In fact, the variance of the factor boosts from the first to the second period for agricultural raw material, cereals, metals and vegetable oil and protein meal, which increased 101%, 104%, 230% and 87%, respectively. In contrast, for

beverages and meat and seafood, variances decreased by 47% and 41%, respectively, from first period to the second.

The FAVAR model results

In this section, we present the estimation results of the FAVAR model for the two periods analysed. Our purpose is to examine the short-run linkages among the variables considered in the study. In particular, we are interested in analysing the impact of uncertainty and macroeconomic and speculative variable shocks on nonfuel commodity prices. We use impulse response functions, as they can trace over time the effects on a variable of an exogenous shock from another variable. Before starting with the results, some final specifications of the model are briefly discussed.

As suggested by Bermanke *et al.* (2005), we use all variable data in first differences in order to induce stationarity. Moreover, all price series have been transformed

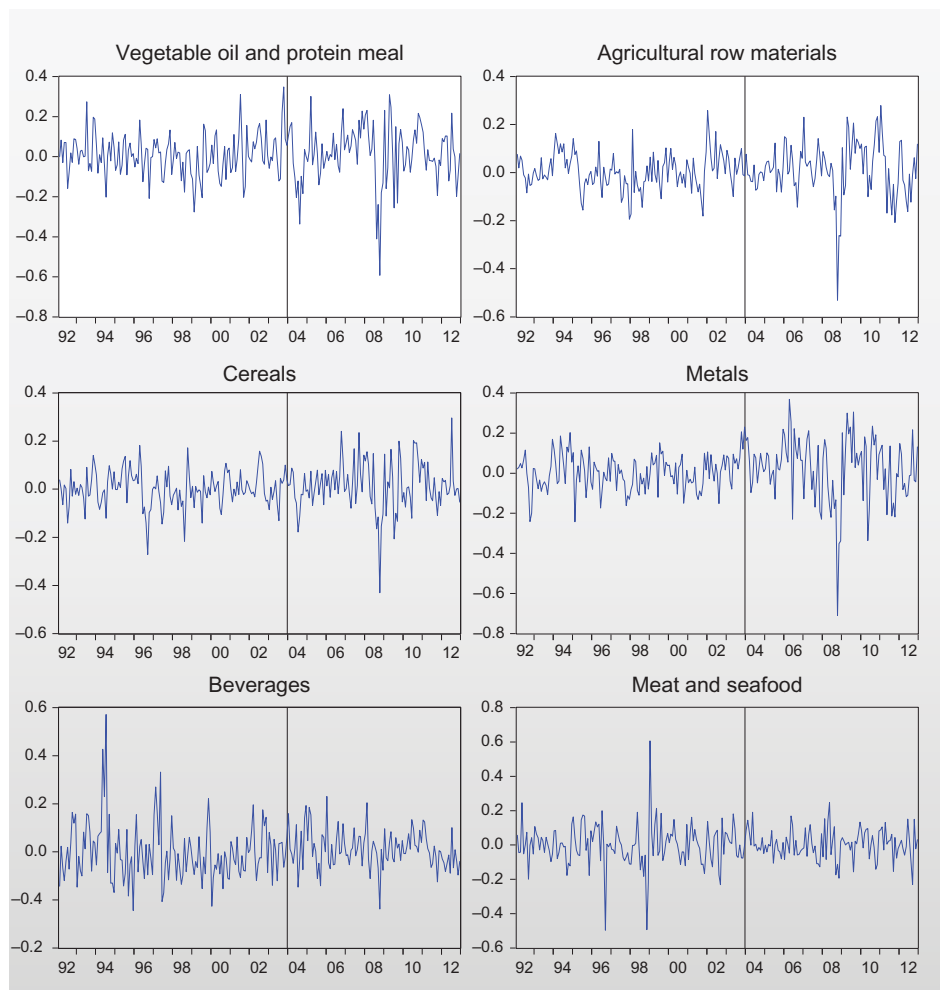


Fig. 4. Common factor by category estimated through Kalman filter

Note: DFMs by category are estimated in the state-space using the Kalman filter. The factor follows an AR(1) process.

into real prices by dividing them by the US consumer price index. We use generalized impulse responses for the impulse response function as described by Pesaran and Shin (1998), whose approach does not depend on the VAR ordering.

Our main results in both sub-periods are shown in Figs 5 and 6. Each figure shows the impulse responses of the nonfuel commodity factor to a one-SD shock in all of the macroeconomic and financial variables selected. The FAVAR models, within the first period (pre-

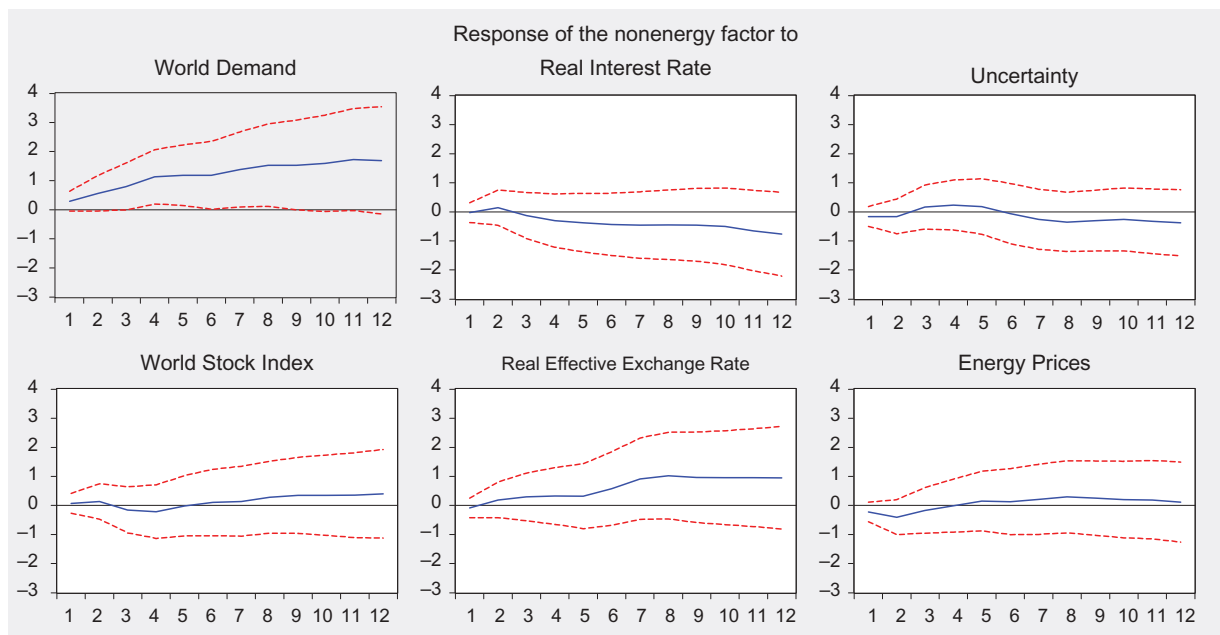


Fig. 5. Accumulated responses of nonfuel commodity prices to shocks in macroeconomic and financial variables. Period 1992:2/2003:11

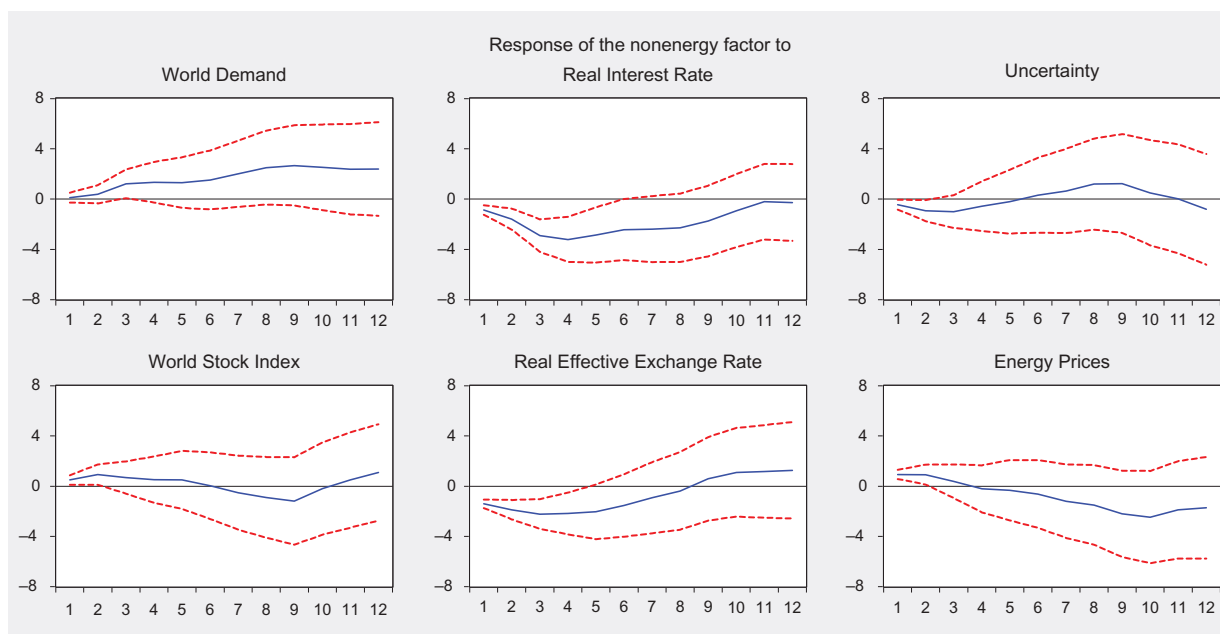


Fig. 6. Accumulated responses of nonfuel commodity prices to shocks in macroeconomic and financial variables. Period 2003:12/2012:12

December 2003) and the second period (post-December 2003), were estimated with five and nine lags, respectively, in order to account for residual autocorrelation. The properties of the residuals of the estimated models have been analysed and show the adequacy of the estimated models. For details see Appendix 3.²

For the period prior to December 2003, the impulse response functions of Fig. 5 show that the comovement in nonenergy commodities is only affected by a world demand shock.

We obtain the following results for the second period: a world demand shock seems to show a significant positive rise in nonenergy commodity prices. Other macroeconomic variables, such as the real interest rate and US real effective exchange rates, have a significant negative impact on nonfuel commodity prices. The negative effect of a shock on real interest rate on the nonenergy factor lasts for 5 months, one of the more lasting effects regarding the impulse response results of the FAVAR model. On the other hand, a real appreciation of the US dollar makes commodity prices fall, as most of the raw material prices are internationally traded and quoted in US dollars. The negative effect takes 4 months to vanish.

A supply shock, proxied in this study by energy prices, has an immediate effect on the nonfuel factor. However, this effect tends to disappear after the second month. In this context, we agree with the idea that there is a spillover effect from energy to nonenergy commodity prices, enhanced perhaps by the financialization of commodity markets. This effect, however, is less lasting than the shocks in fundamentals such as the real exchange rate and the real interest rate.

With regard to financial variables, we find significant effects of nonfuel commodity prices. Impulse response estimates indicate that a shock in the world stock index is followed by a 2-month rise in nonfuel commodity prices. These results show the strengthening of cross-market linkage after late 2003.

In relation with the effects of uncertainty, we find that an increase in this variable leads to a decrease in the nonenergy commodity prices for 2 months. It seems that uncertainty or volatility in the financial market makes investors more risk averse.

Variance decompositions

Now, we present the variance decomposition of the nonenergy factor for both sub-periods in order to assess the relative importance of each of the variables in determining the nonenergy price comovement. Yet, variance decomposition requires orthogonality of the inputs if we require

the sum of variance due to each component to total 100%. We rely on a given Cholesky decomposition and, therefore, we propose the following ordering of the variables in the VAR: in both of the sub-periods we place our nonenergy factor last, based on the assumption that nonenergy commodity market shocks have no contemporaneous effects on macroeconomic variables. World demand, proxied in our study by global industrial production, is placed first, followed by the interest rate, the VIX, the world stock index, real effective exchange rate and the IMF energy index. This implies that demand shocks instantaneously affect all equations of the system, which seems logical if we consider that the real exchange rates and interest rates respond greatly to macroeconomic conditions.³ Additionally, placing the nonenergy commodity factor last ensures that the estimated impact on the last variable, which is the one we focus on, is netted out of the joint impact of financial and macroeconomic variables. Although the relative ordering of VIX, the world stock index and real exchange rate is unclear, we find that the impulse response results are robust to the ordering of these variables.

Figure 7 illustrates the percentage of variance of the forecasting error of the nonenergy factor, at a 12-month horizon, which is attributable to a shock in each of the macroeconomic and financial variables of the model.

In the first sub-period, the nonenergy factor is the most important driver of its own prices, accounting for more than 70% of price fluctuation. Therefore, inertia has a dominant role in explaining the factor fluctuations. Considering the other variables rather than the lags themselves of the common factor, world demand shocks are the most important driver, explaining up to 7% of nonenergy price changes. The rest of the variable shocks account for less than 5% of the nonenergy factor.

In the second sub-period, the relative importance of the shocks is greatly altered. Real interest rate is now the most important driver of comovements in nonenergy commodity prices, accounting for up to 26% of price fluctuations. The role of a shock in uncertainty is notable, explaining 17.4% of the variation in nonenergy prices at all horizons, the second most important driver of the nonenergy factor for this period. Shocks in the fundamentals, such as real effective exchange rate and world demand, account for up to 17% and 8%, respectively. Finally, the world stock index variable and the energy variable explain 8% and 7%, respectively, of the variation in nonenergy price fluctuations.

The most important result of this comparison is the increase in the role of uncertainty to explain nonenergy fluctuations. The fact that the uncertainty shock accounts for a larger share of variance decomposition of nonenergy

² A dummy on December 2008 has been included in the FAVAR model for the second period to take into account the world economic crisis. We did not detect heteroscedasticity in the residuals of the FAVAR model.

³ As a robustness check, we evaluated a different Cholesky ordering of the variables in the VAR with similar results.

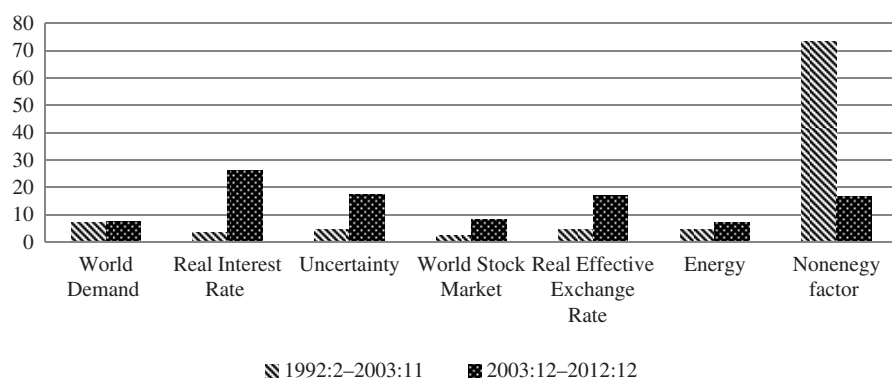


Fig. 7. Variance decomposition of the nonenergy factor

Note: Variance decomposition at horizon 12.

commodity prices than fundamentals, such as real exchange rate and world demand, emphasizes its importance in determining nonenergy spot price formation. Financialization of commodities may make investors much more aware of short-term economic developments, and may cause passive investors to take collective decisions such as selling risky assets, such as commodities, when uncertainty increases, given their correlation with other risky assets.

In conclusion, in the short term, the speculative variables used in our study, which reflect the financial market condition (world stock index) and uncertainty (VIX), significantly affect the real prices of nonfuel commodities in the second period, after the end of 2003.

IV. Conclusions

This article improves the understanding of the comovement among nonfuel commodity prices in the short run, adding to the literature on financialization of commodity markets. First, we evaluate the magnitude in the comovement of 44 monthly nonfuel commodity prices from the period February 1992 to December 2012. Second, we split our sample in December 2003, as it is the starting date of an unprecedented upsurge of investment into commodity trade funds, in an attempt to ascertain whether comovement is affected in the same way by fundamentals, financial and speculative variables. We evaluate the responses of nonenergy commodities to macroeconomic and financial shocks for these two sub-periods. As we focus on the short run, special attention is paid on the role of the world stock index variable as well as uncertainty in the financial market as a possible determinant of commonalities in commodity prices. With regard to the methodology, we use the FAVAR approach of Bernanke *et al.* (2005) in order to check that which drives comovement in nonenergy commodity prices.

Our results highlight the importance of comovement among nonfuel commodities that began after the financialization of commodities in late 2003. In this period, we found not only an increase in the variance explained by the factor attributable to the comovement between raw materials, but also that variables such as uncertainty and the world stock index significantly impact the factor which relates comovements in nonenergy commodity prices. Moreover, the variance decomposition analysis shows that uncertainty element plays a more important role in explaining nonenergy fluctuation than fundamentals such as the real exchange rate and the real interest rate.

In short, although classical macroeconomic factors of supply and demand are able to explain many of the sharp movements in the short term in nonfuel commodity prices, the growing importance of uncertainty as an important determinant of commonalities in nonenergy commodities is an element that cannot be ignored.

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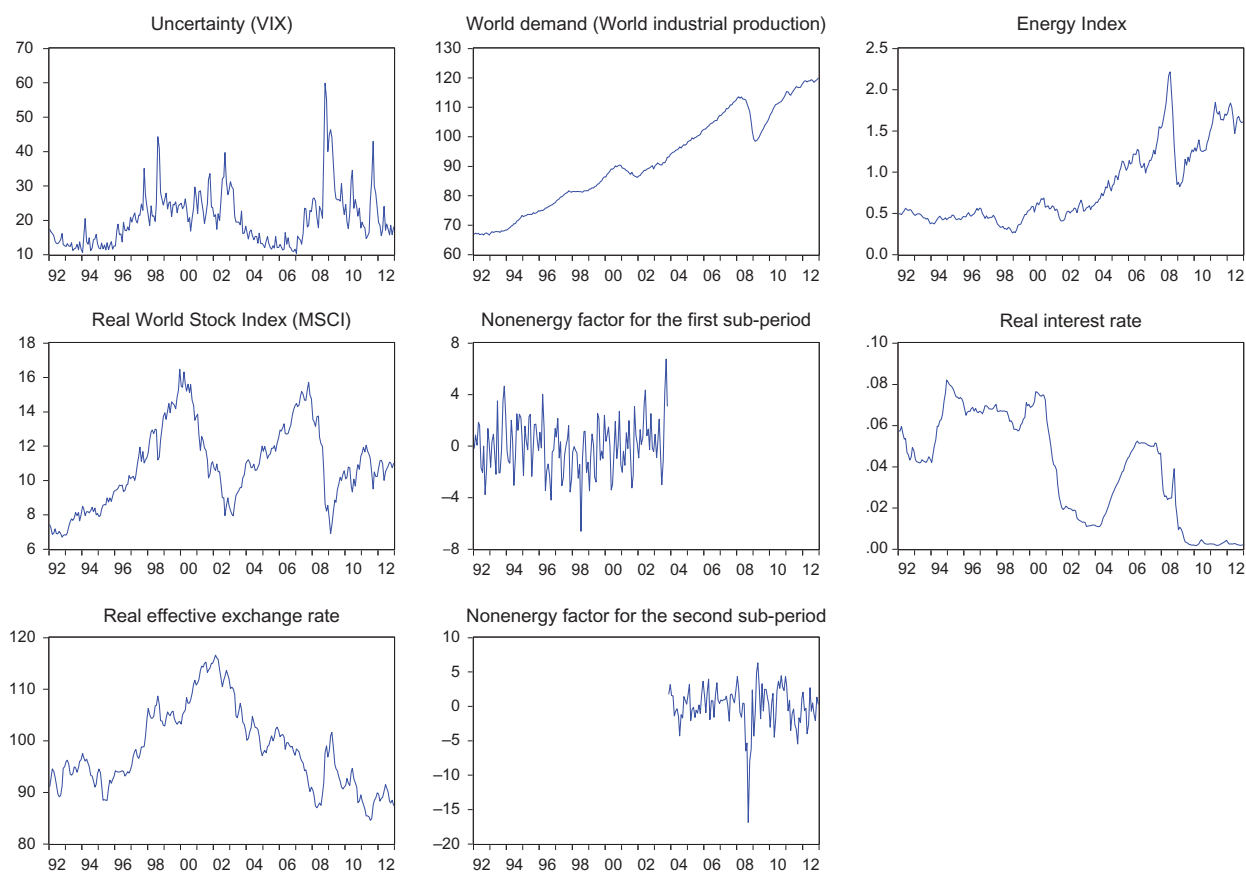
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Appendix 1. Nonfuel commodities

IMF category		Commodity	
Edibles	Cereals	Rice	Maize
		Barley	Wheat
	Meat and seafood	Beef	Poultry (chicken)
		Lamb	Fish (salmon)
		Swine (pork)	Shrimp
	Beverages	Cocoa beans	Coffee, Robusta
		Coffee, other mild Arabicas,	Tea
	Vegetable oils and protein meals	Soybeans	Sunflower oil
		Soybean meal	Olive oil
		Soybean oil	Groundnuts (peanuts)
		Palm oil	Rapeseed oil, crude
		Fishmeal	
	Other Edibles	Sugar EU	Oranges
		Sugar US	Hard logs
		Bananas	
Industrial inputs	Agricultural raw materials	Soft logs	Wool, fine
		Hard sawnwood	Rubber
		Soft sawnwood	Hides
		Cotton	Wool, coarse
		Aluminium	Zinc
	Metals	Copper	Nickel
		Iron ore	Lead
		Tin	Uranium

Appendix 2. Graphs of the macroeconomic and financial variables

Appendix 3. Residual autocorrelation analysis of the FAVAR model

	Test	Chi ²	<i>p</i> -value
1992:2–2003:11	LM(1)	$x^2(49) = 44.49$	0.6563
	LM(2)	$x^2(49) = 43.09$	0.7101
	LM(3)	$x^2(49) = 51.92$	0.3605
	LM(4)	$x^2(49) = 44.26$	0.6651
	LM(5)	$x^2(49) = 47.06$	0.5517
	LM(6)	$x^2(49) = 39.49$	0.8318
	LM(7)	$x^2(49) = 48.78$	0.4820
	LM(8)	$x^2(49) = 38.42$	0.8616
	LM(9)	$x^2(49) = 50.33$	0.4099
	LM(10)	$x^2(49) = 55.48$	0.2436
	LM(11)	$x^2(49) = 50.33$	0.4202
	LM(12)	$x^2(49) = 35.30$	0.9292
2003:12–2012:11	LM(1)	$x^2(49) = 55.26$	0.2501
	LM(2)	$x^2(49) = 47.01$	0.5539
	LM(3)	$x^2(49) = 46.43$	0.5778
	LM(4)	$x^2(49) = 50.03$	0.4319
	LM(5)	$x^2(49) = 60.54$	0.1247
	LM(6)	$x^2(49) = 49.22$	0.4641
	LM(7)	$x^2(49) = 39.56$	0.8297
	LM(8)	$x^2(49) = 56.75$	0.2085
	LM(9)	$x^2(49) = 52.42$	0.3425
	LM(10)	$x^2(49) = 43.87$	0.6824
	LM(11)	$x^2(49) = 55.29$	0.2490
	LM(12)	$x^2(49) = 46.67$	0.5694

Note: LM = Lagrange multiplier test.